TEXORA · LECTURE SHEET

Cost Elements: Materials, Labour and Overheads

The three elements of cost in depth: material control and inventory levels with the EOQ worked in full, labour cost and incentive schemes (Halsey and Rowan) with worked wages, and the collection, apportionment and absorption of overheads.

AC-4205 · COST AND MANAGEMENT ACCOUNTING · TOPIC 3 · LEVEL 4 TERM 2

1. The elements and how cost flows

Every product cost is built from three elements — **materials, labour and expenses** — each split into direct and indirect. Direct parts are traced to units; indirect parts gather as overheads and are absorbed. The flow through a factory's accounts:

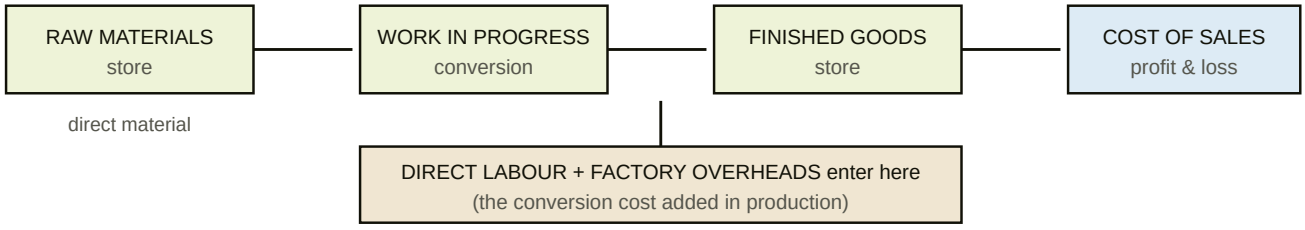


Figure 1 — The cost flow. Materials move store → WIP → finished goods → cost of sales; labour and factory overhead join at the WIP stage. Administration and selling overheads are added after production cost (Topic 6).

2. Materials: control and the economic order quantity

Material is usually the largest cost element in textiles, so its control system matters: purchase requisition and competitive ordering, goods-received inspection, bin-card and stores-ledger records, issue only against requisitions, and perpetual inventory with cycle counts. Stock levels are set by policy: **reorder level** (when to order), **minimum level** (safety stock), **maximum level** (capital discipline) — and the order size by the **Economic Order Quantity**, which balances ordering cost against holding cost:

$$EOQ = \sqrt{(2 \times D \times S \div H)}$$

where D = annual demand, S = cost per order, H = holding cost per unit per year. At the EOQ, annual ordering cost and annual holding cost are equal, and total inventory cost is minimised.

WORKED EXAMPLE — EOQ FOR A DYE CHEMICAL

Annual demand D = 36,000 kg; cost per order S = Tk 1,500; holding cost H = Tk 40 per kg per year.

EOQ = √(2 × 36,000 × 1,500 ÷ 40) = √2,700,000 ≈ **1,643 kg per order**.

Orders per year = 36,000 ÷ 1,643 ≈ **21.9 orders**.

Annual ordering cost ≈ 21.9 × 1,500 ≈ Tk 32,863; annual holding cost = (1,643 ÷ 2) × 40 ≈ Tk 32,863 — equal, as the theory requires.

Ordering much more (say 6,000 kg lots) would slash ordering cost but pile up holding cost and expiry risk; ordering much less reverses the error. EOQ finds the balance.

Issues from store are priced (FIFO, or weighted average) so that jobs are charged consistently; material losses are classified as **normal** (inherent, e.g. evaporation, absorbed by good output) or **abnormal** (avoidable, charged out as a loss — the same logic Topic 4 applies in process costing).

3. Labour: cost, idle time and incentive wages

Labour cost is captured by time records (clock/gate records for attendance, job cards for time on work). Pay splits into the wage for productive time, **overtime premium** (the extra over the basic rate — usually charged to overhead unless a specific job caused it), and **idle time** (paid time without production — machine breakdown, waiting for material; abnormal idle time is a loss, normal idle time an overhead). Where output can be measured, **incentive schemes** share the gain from working faster than standard:

WORKED EXAMPLE — HALSEY AND ROWAN PREMIUM PLANS

A worker is allowed 8 hours for a job (standard time) and completes it in 6 hours (time taken). Rate Tk 100 per hour. Time saved = 2 hours.

Time wages (guaranteed floor) = 6 × 100 = Tk 600.

Halsey plan (worker gets 50 % of the value of time saved):

Wage = (6 × 100) + 50 % × (2 × 100) = 600 + 100 = **Tk 700**.

Rowan plan (bonus is the proportion time saved ÷ time allowed of the time wages):

Wage = 600 + (2 ÷ 8) × 600 = 600 + 150 = **Tk 750**.

Reading the numbers: both plans pay more than time wages (the incentive) but less than straight piece rates would at double speed; Rowan pays more than Halsey when a small fraction of time is saved, and its bonus tapers as time saved grows — protecting the employer against loose standards, which is why the two plans coexist.

Employer's total labour cost also includes contributions and benefits beyond the wage packet; costing uses the full cost per hour, not the take-home pay.

4. Overheads: from collection to absorption

Indirect costs are handled in four steps. **Collection and allocation:** each overhead is charged to the cost centre that caused it where possible (machine insurance to that machine's centre). **Apportionment:** shared costs are spread on fair bases — rent by floor area, power by metered use or machine capacity, canteen by headcount. **Re-apportionment:** service centres (maintenance, stores, boiler house) are then spread over production centres. **Absorption:** each production centre's total overhead is charged to units through a rate:

$$\text{Overhead absorption rate} = \text{Budgeted overhead} \div \text{Budgeted base (hours or units)}$$

WORKED EXAMPLE — MACHINE HOUR RATE AND LABOUR HOUR RATE

A sewing department budgets factory overhead of Tk 18,00,000 for 60,000 machine hours.
Machine hour rate = 18,00,000 ÷ 60,000 = **Tk 30 per machine hour**.
A job using 45 machine hours absorbs 45 × 30 = **Tk 1,350** of overhead into its cost.
Measured on labour instead: if the same overhead were recovered over 90,000 direct labour hours, the direct labour hour rate = 18,00,000 ÷ 90,000 = **Tk 20 per labour hour**.
Choice of base follows cause: machine-paced work absorbs by machine hours; hand work by labour hours; other common bases are units of output, direct wages percentage and prime cost percentage.

Because rates are set from budgets before the year runs, actual overhead and actual activity rarely match the budget: the difference is **under- or over-absorption**, analysed at period end and written off or carried per policy — the standard-costing treatment appears in Topic 9.

Glossary

EOQ — economic order quantity minimising ordering + holding cost. **Reorder/minimum/maximum levels** — stock policy thresholds. **Idle time** — paid but unproductive time. **Overtime premium** — the excess of overtime pay over the basic rate. **Halsey / Rowan plans** — premium bonus schemes sharing time saved 50:50 / proportionally. **Apportionment** — spreading a shared cost over centres on a fair basis. **Absorption** — charging overhead to units via a rate on a chosen base. **Machine hour rate** — overhead per machine hour worked.

Key points

• Materials flow store → WIP → finished goods; labour and factory overhead join at WIP as conversion cost.
• EOQ = √(2DS/H): for D 36,000, S Tk 1,500, H Tk 40 → 1,643 kg, ~21.9 orders a year, ordering ≈ holding ≈ Tk 32,863.
• Incentives: 6 h taken against 8 h allowed at Tk 100/h pays Tk 600 time wages, Tk 700 under Halsey, Tk 750 under Rowan.
• Overheads are allocated, apportioned, re-apportioned and absorbed — Tk 18,00,000 over 60,000 machine hours gives Tk 30/MH, so a 45-hour job carries Tk 1,350.

Practice

1. Compute the EOQ for annual demand 24,000 units, ordering cost Tk 1,200 per order, holding cost Tk 30 per unit per year. How many orders per year does it imply?
2. A job's standard time is 10 hours; a worker finishes in 8 hours at Tk 120 per hour. Compute the wage under (a) time rate, (b) Halsey, (c) Rowan.
3. A department budgets overhead Tk 9,60,000 and 48,000 direct labour hours. Find the absorption rate and the overhead charged to a job taking 120 labour hours.
4. Explain why abnormal idle time is excluded from product cost while normal idle time is included.

Answer key

1. EOQ = √(2 × 24,000 × 1,200 ÷ 30) = √1,920,000 ≈ **1,386 units**. Orders = 24,000 ÷ 1,386 ≈ **17.3 orders per year**.
2. Time saved = 2 h. (a) 8 × 120 = **Tk 960**. (b) Halsey: 960 + 50 % × (2 × 120) = 960 + 120 = **Tk 1,080**. (c) Rowan: 960 + (2 ÷ 10) × 960 = 960 + 192 = **Tk 1,152**.
3. Rate = 9,60,000 ÷ 48,000 = **Tk 20 per labour hour**. Job charge = 120 × 20 = **Tk 2,400**.
4. Product cost should measure the resources a unit ought to consume. Normal idle time (setting up, brief waits built into the process) is an unavoidable part of production, so it is absorbed as overhead by all units. Abnormal idle time (breakdown through negligence, a preventable material stock-out) is a loss of the period: loading it into unit costs would hide the failure inside product costs and penalise good production for a controllable breakdown.